

Trevor Keller, Ph. D.

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Scientist and Sysadmin thriving at the intersection of mathematical models, high-performance computing, and data storage. Specializing in Linux clusters and solid-state microstructure evolution in metals. Passionate about building a better tomorrow.

EXPERIENCE

NATIONAL INSTITUTE OF STANDARDS & TECHNOLOGY | ENGINEER

2015 – Present | Gaithersburg, Maryland

- Co-admin with root on a 100-node cluster serving 70 scientists with 300 TB of replicated working data.
- Built and tuned VASP on behalf of users, managed access via LDAP.
- Wrote a widely cited paper with 12 colleagues explaining commonly observed microstructure of additively manufactured metals.

DAYSTAR TECHNOLOGIES | PROCESS ENGINEER

2006 – 2009 | Clifton Park, New York

- Evaluated aqueous ZnO deposition as a non-toxic substitute for CdS in thin-film solar cells. Could not match the heavy metal's performance.
- Achieved 72× scale-up in aqueous CdS deposition area while generating only 6× the waste using a flow-through reactor geometry.
- Assessed a dry commercial reactor in Helsinki, Finland, reporting on its viability to senior management.

PROJECTS

HPC CARPENTRY | STEERING COMMITTEE MEMBER

2019 – Present | hpc-carpentry.org

- Collaborated with a worldwide group of volunteers, mostly cluster sysadmins, to develop an approachable two-day introduction to high-performance computing for novices.
- Successfully drove our petition for official recognition from The Carpentries, now working through their Incubation program.
- Codified asynchronous democratic decisionmaking practices.

PFHUB

2015 – Present | pages.nist.gov/pfhub

- Collaborated with North American academics and Federal research staff at twice-yearly workshops on phase-field methods.
- Assisted with design and implementation of a website to ingest and display submitted results for comparison, without CMS.
- Collaborating with a half-dozen researchers to build a FAIR schema for phase-field simulation datasets.

MMSP 2012 – Present | github.com/mesoscale

- Executed parallel jobs with 8,192 cores on a BlueGene/Q supercomputer.
- Refactored templated C++ codebase to use non-blocking MPI calls.
- Implemented a staged file I/O method to resolve corruption on disk.
- Invisibly swapped endianness of binary data at read-time due to mismatch between supercomputer (big) and workstation (little).

NOTABLE WORKS

HPC Carpentry: doi.org/10.22369/issn.2153-4136/15/1/6
PFHub platform: doi.org/10.5334/jors.276
Additive research: doi.org/10.1016/j.actamat.2017.05.003

SKILLS

OPERATIONS | DEBIAN

Proficient:

SSH • Slurm • Bash • Ansible • YAML

Experienced:

LDAP • Grafana • NFS • mdadm

PROGRAMMING

Proficient:

Python • git • C/C++ • Make • MPI

Experienced:

Pandas • R • CUDA • Apptainer

DOCUMENTATION

Markdown • L^AT_EX • Doxygen • Pandoc

LIBRARIES/Frameworks

Lmod • MPI • OpenMP • Jekyll

EDUCATION

RENSSELAER POLYTECHNIC INSTITUTE

Ph.D., MATERIALS ENGINEERING

2011 – 2015 | Troy, New York

📖 Bias in Polycrystal Topology Caused by Grain Boundary Motion by Mean Curvature

M.S., MATERIALS ENGINEERING

2009 – 2011 | Troy, New York

📖 Effects of Magnesium(II) on Zinc Oxide Nanorod Growth From Aqueous Solution

B.S., CHEMICAL ENGINEERING

2002 – 2006 | Troy, New York

AWARDS

2021 Dept. of Commerce Bronze Medal
2018 NIST MML Accolade